GenMedX

AI Healthcare Agent

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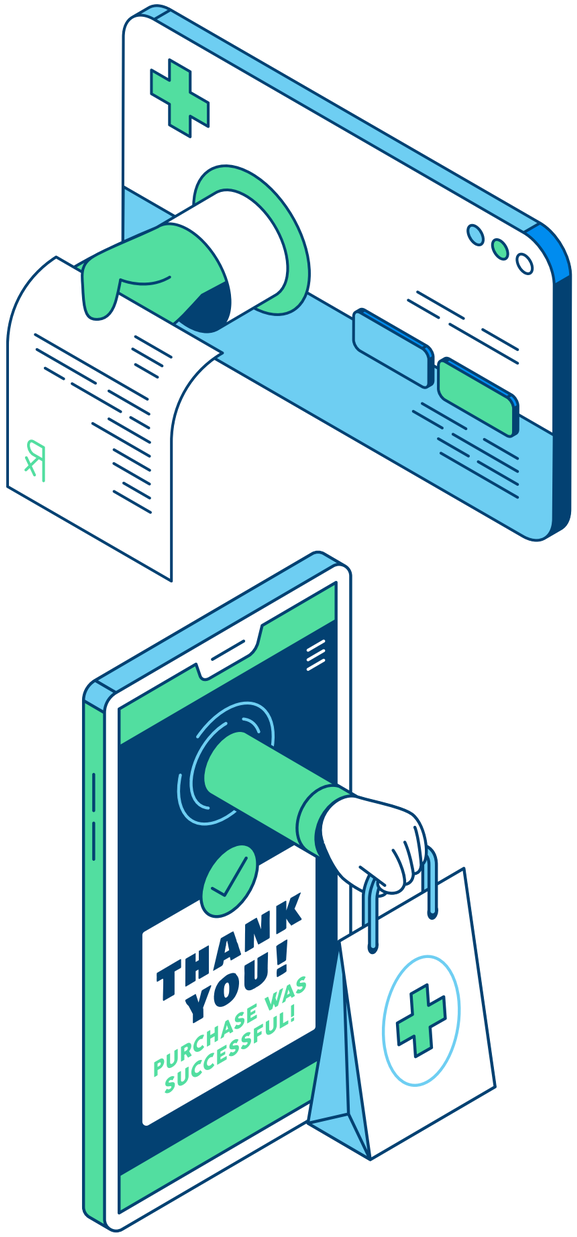


EXECUTIVE SUMMARY

GenMedX is an **AI-powered healthcare agent** designed to transform structured Emergency Department (ED) data into natural-language summaries while providing clinical decision support through diagnosis suggestions, medical Q&A, and patient risk assessment all integrated within a single intelligent, cloud-based system.

Our system integrates machine learning and natural language processing using Transformer-based models and large language models (LLMs) to deliver intelligent, context-aware healthcare assistance.

4 Core Capabilities:

- The **ED Visit Summarizer Module,** which generates discharge-ready summaries.
- The **Symptom-to-Diagnosis Module** offers differential diagnosis suggestions.
- The **Medical Q&A Module** answers domain-specific questions using curated evidence.
- And the **Risk Prediction and Tests Recommendation Module**, which estimates patient risk scores and recommends follow-up tests.

Demonstrates how AI can make healthcare communication more transparent, reduce documentation burden, and empower both clinicians and patients through data-driven understanding and decision support.

Data Pipeline & System Design

Data Ingestion & Storage

- Patient data ingested from **Google Drive** → **Google Cloud Storage** → **BigQuery** **via** **Cloud Composer**. Structured data curated into BigQuery Views for downstream tasks.

Inference Layer (LangGraph Framework)

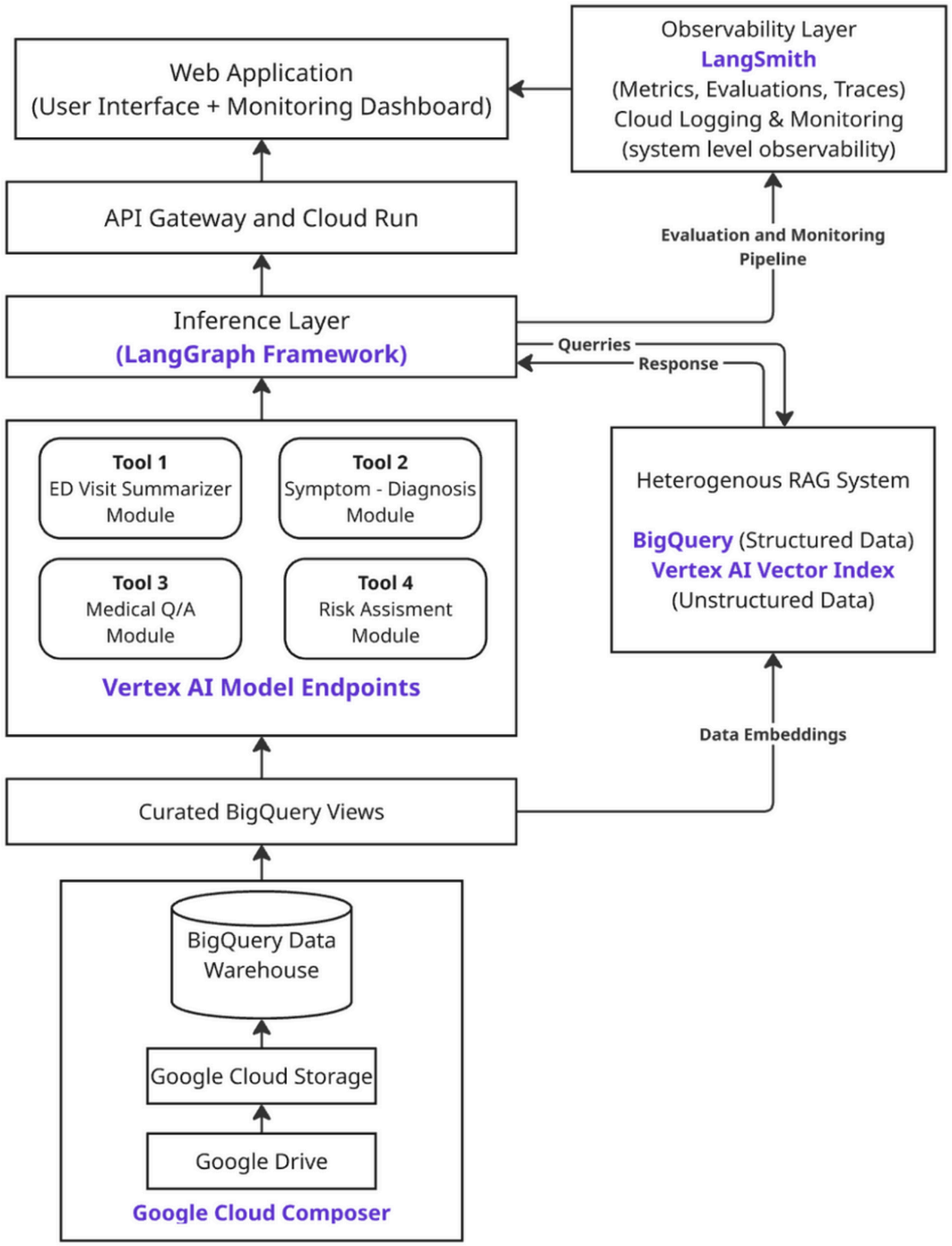
- Orchestrates model calls as functions (tools). Each tool is deployed as a **Vertex AI** Model Endpoint.

Heterogeneous RAG (Retrieval-Augmented Generation) System

- **BigQuery** for structured clinical data.
- **Vertex AI Vector Index** for unstructured notes/documents.

Observability & Monitoring

- **LangSmith** for trace-level evaluations, metrics, and error analysis.
- **Cloud Logging** & Monitoring for system health and performance.
- Integrated evaluation pipeline to track model reliability, factuality, and patient-safety metrics.



Tool 1: ED Visits Summarizer

DATASET-1

The **MIMIC-IV-ED (v2.2)** dataset is a large-scale, publicly available resource that contains detailed records of approximately **425,000 emergency department (ED) visits** that took place at the Beth Israel Deaconess Medical Center (BIDMC) between 2011 and 2019. It is part of the broader MIMIC-IV data ecosystem and is **hosted on PhysioNet**, a platform for freely sharing medical data for research.

Table Name	Description	Record Count
edstays	Emergency department visits (unique stays IDs)	~425,087
triage	Triage assessment, vitalsigns, chief complaints	~425,087
vitalsign	Vital Signs recorded throughout the ED stay	~1,564,610
diagnosis	Discharge diagnosis with ICD-10 codes	~899,050
medrecon	Medication reconciliation at admission	~2,987,342
pyxis	Administered medications during ED stays	~1,586,053

Tool 1: ED Visits Summarizer

Models Used

- **T5-small (fine-tuned):** Abstractive model that turns each ED stay's rows into a clear, second-person paragraph.
- **LangGraph + pandas (orchestration):** Loads CSV, groups by stay_id, calls the ML model per stay, joins results.
- **Serving:** Docker container on Vertex AI Endpoint (/summarize) with a Cloud Run web UI.

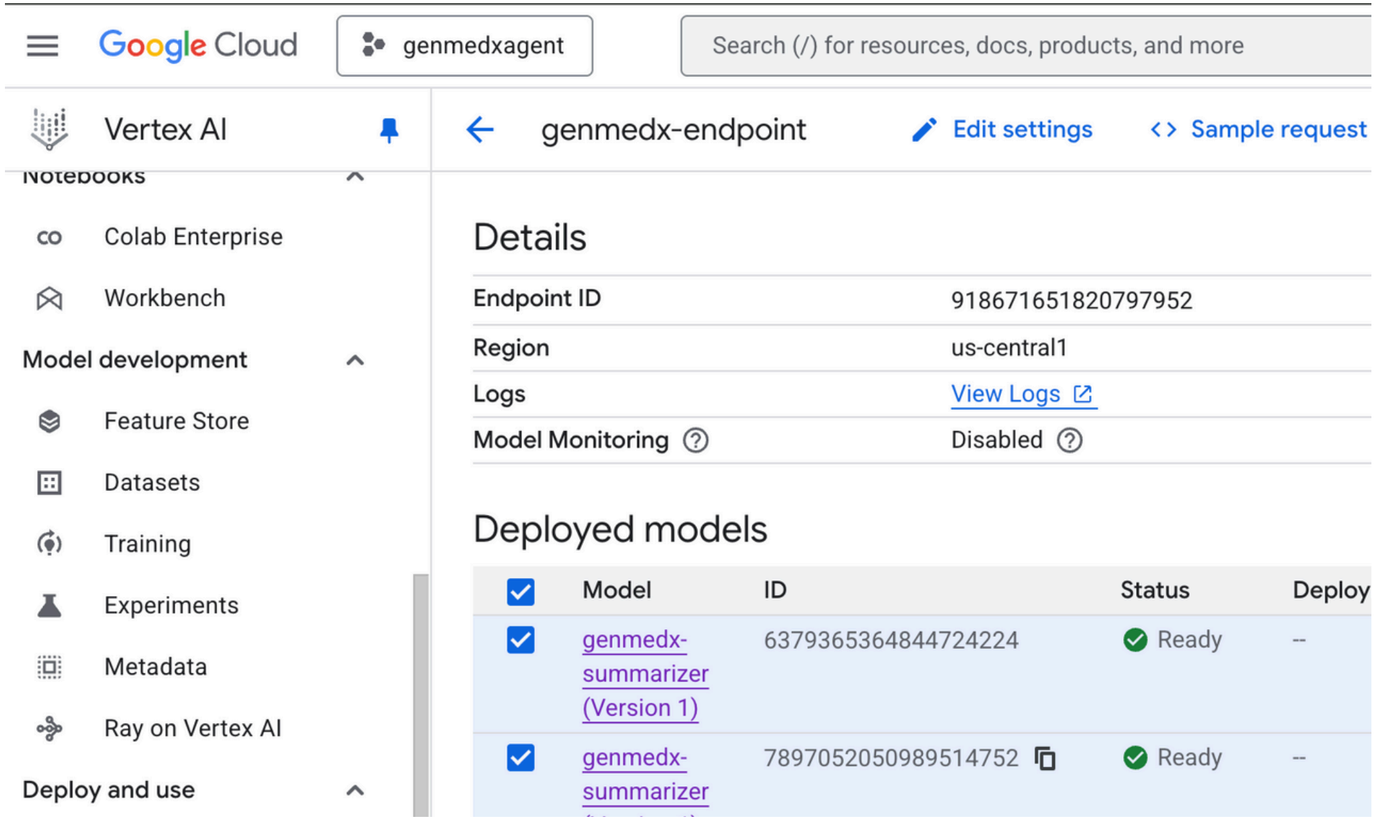
Summarization Workflow

- **Web UI (Cloud Run)** → user enters Patient ID
- **Flask:** container (Vertex Model Endpoint) calls agent.py
- **Agent:** groups rows by stay_id and calls T5-small to summarize each stay
- **Filter**: Select rows for the input subject_id; group by stay_id to isolate each visit.
- **Assemble**: Concatenate paragraphs → final patient history.
- **Generate**: T5-small produces one concise paragraph per stay.
- **API**: POST /summarize - {"instances":[{"patient_id":"14601430","question":"Summarize my hospital stay history?"}]}
- **Deployment**: Docker → Artifact Registry → Vertex AI Model & Endpoint (us-central1); demo UI on Cloud Run with a service account (ADC) calling the endpoint.

Tool 1: ED Visits Summarizer

Input:
Input is patient_id (string/int) plus a free-text question.

Output:
list of short paragraphs, one per stay, joined for convenience




```
gcloud ai endpoints predict ${ENDPOINT_ID} \
  --project=${PROJECT_ID} \
  --region=${REGION} \
  --json-request=request.json
Using endpoint [https://us-central1-prediction-aiplatform.googleapis.com/]
['Stay on 2167-02-26:\nStay on 2167-02-26: You arrived as a walk-in with abdominal pain, n/v. Triage vitals were T 99°F; HR
 110/min; RR 18/min); SpO2 95%; BP 120/90 mmHg. You were admitted. The primary diagnosis was Nausea with vomiting, unspecif
ied.']
```

Tool 1: ED Visits Summarizer

Training script: train_t5.py (Transformers Trainer).

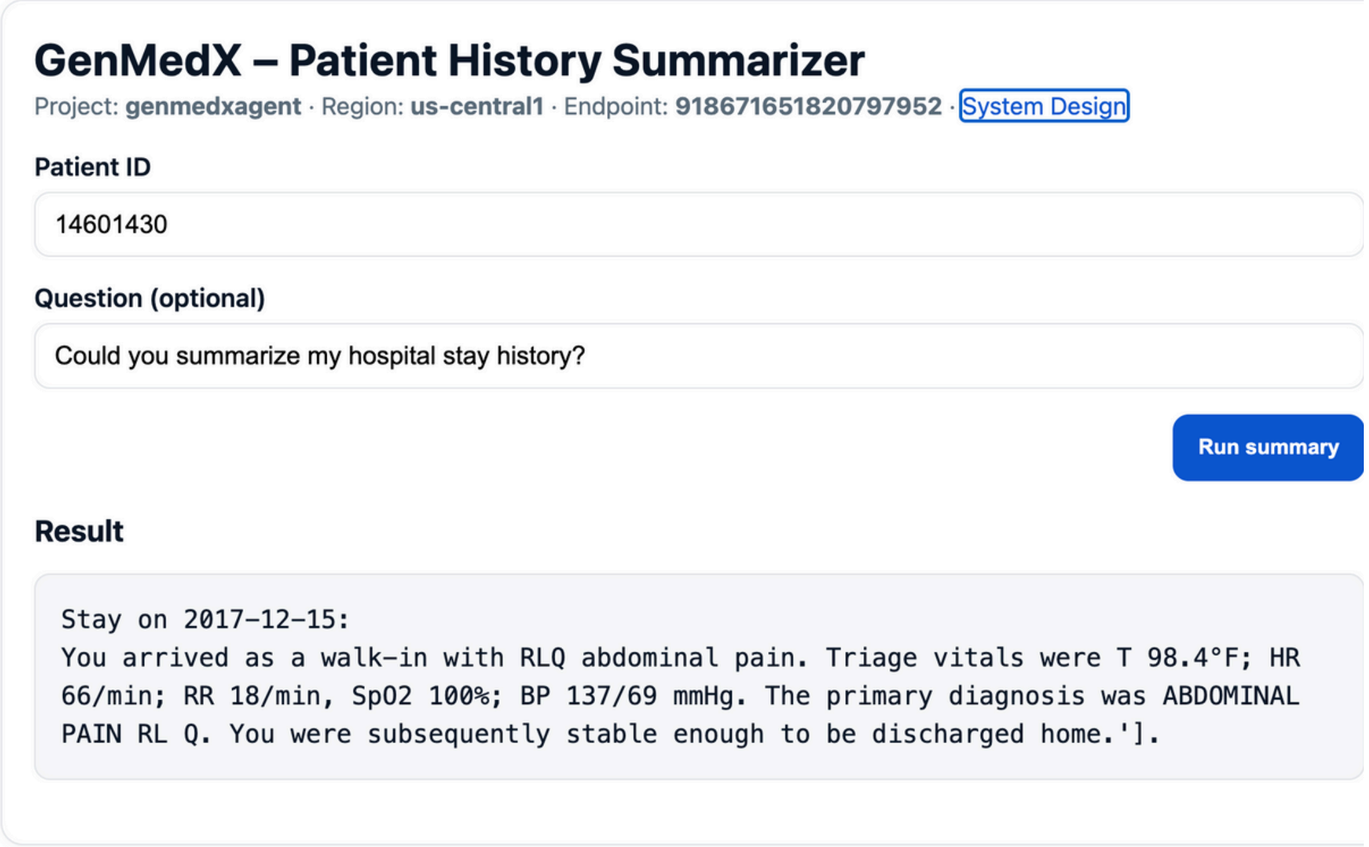
- 15 epochs, batch size 4, LR 5e-5, cmax source 256, max target 160, seed 42.
- **Data**: data/train.jsonl, data/val.jsonl (input_text → target_text pairs).
- **Metric**: ROUGE on decoded outputs.

Evaluation (validation):

- Loss: 0.1827
- ROUGE-1: 93.66 ROUGE-2: 90.75
- ROUGE-L: 93.64 (Lsum 93.34)

Ops & Demo:

- **Endpoint**: genmedx-endpoint (us-central1).
- **Test**: gcloud ai endpoints predict
- ENDPOINT_ID --region us-central1 --json-request



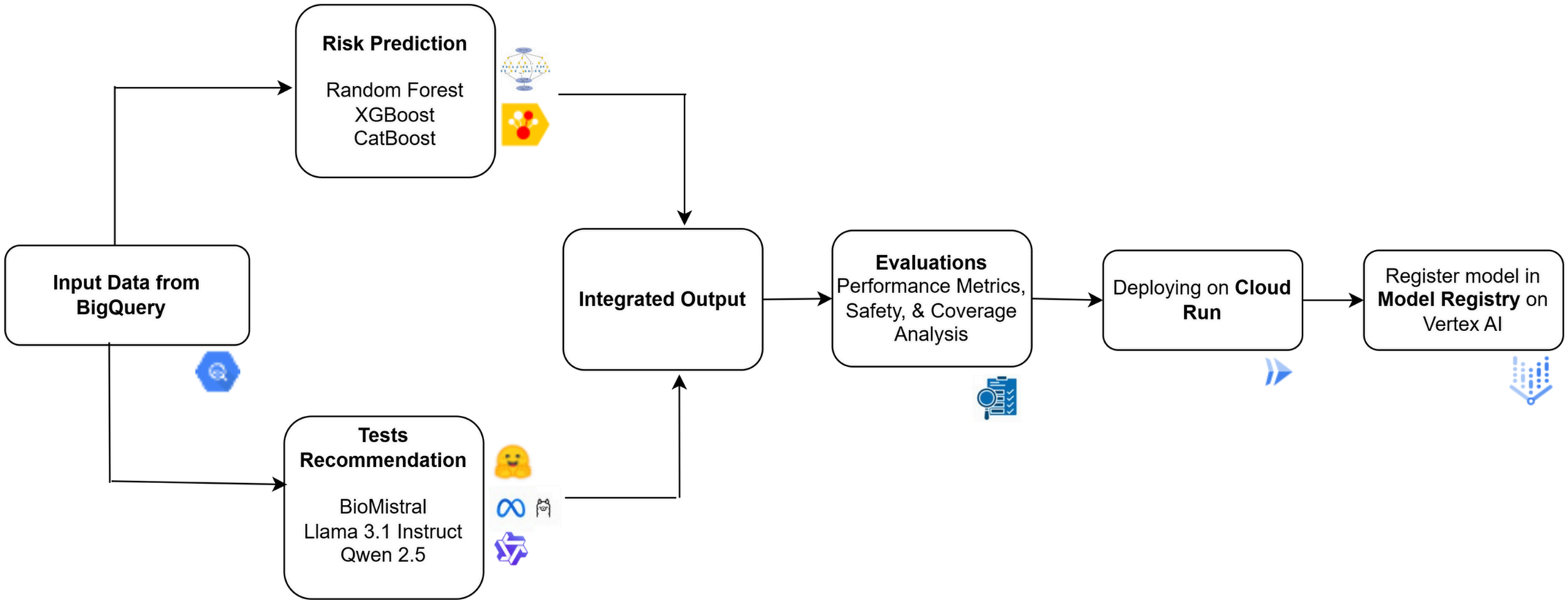
Tool 2: Risk Prediction & Tests Recommendation Module

The data used comes from the MIMIC-IV ED database. For this module, we extracted the triage table, which includes patient vitals, chief complaints, and acuity scores. These serve as the foundation for both risk prediction and test recommendation tasks.

Dataset attributes:

- Chief complaint (free-text),
- SBP (mmHg) – systolic blood pressure
- DBP (mmHg) – diastolic blood pressure
- O2 saturation (%)
- Heart rate (bpm)
- Respiratory rate (/min)
- Temperature (°C)
- Pain score (0–10)
- Acuity (ED triage acuity level, 1–5)
- Risk label (derived: 1 = High risk, acuity ≤ 2; 0 = Low risk, acuity > 2)

Tool 2: Risk Prediction & Tests Recommendation Module



Tool 2: Risk Prediction & Tests Recommendation Module

1

Random Forest
+
BioMistral

- Ensemble ML for risk
- Domain LLM for test suggestions

2

XGBoost
+
LLama 3.1 Instruct

- Gradient boosting
- Instruction-tuned LLM

3

CatBoost
+
Qwen 2.5
Guarded

- Ordered boosting
- Guard-layered LLM for schema-safe JSON

4

CatBoost
+
Retrieval
Memory(RAG)

- Context retrieval from prior cases
- LLM with augmented reasoning

Tool 2: Risk Prediction & Tests Recommendation Module

cc:	knee pain
sbp:	88
dbp:	66
hr:	77
rr:	22
o2:	55
temp:	37.1
pain:	6


```
{
  "risk": "high",
  "prob": 0.74,
  "rationale": "knee pain + SBP 88 + O2 55",
  "recommended_tests": [
    "ABG/VBG",
    "BMP",
    "Chest X-ray",
    "ECG"
  ],
  "flags": [
    "hypotension",
    "hypoxemia"
  ]
```


Google Cloud | My First Project | Search (/) for resources, docs, products, and more | Search

Open project picker (Ctrl O)

You're working in My First Project

you must turn on 2-step verification or risk

CLOUD SHELL

Terminal (sunlit-hook-455618-n1) × + ▾

Open Editor

The connection to your Google Cloud Shell was lost.

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to sunlit-hook-455618-n1.
Use `gcloud config set project [PROJECT_ID]` to change to a different project.
team1_healthcare_agent@cloudshell:~ (sunlit-hook-455618-n1)$ PROJECT_ID="sunlit-hook-455618-n1"
REGION="us-west2"
BUCKET="sunlit-hook-455618-n1-usw2-model4-archive"
EVAL_DIR="gs://${BUCKET}/eval"
OUT_DIR="${EVAL_DIR}/out"

cat > eval.jsonl <<'EOF'
{"temperature":101.4,"heartrate":128,"resprate":32,"o2sat":88,"sbp":82,"dbp":50,"pain":9,"chiefcomplaint":"shortness of breath and chest tightness","ground_truth":"HIGH_RISK"}
{"temperature":98.0,"heartrate":75,"resprate":16,"o2sat":98,"sbp":120,"dbp":78,"pain":2,"chiefcomplaint":"ankle sprain","ground_truth":"LOW_RISK"}
{"temperature":100.3,"heartrate":105,"resprate":24,"o2sat":92,"sbp":95,"dbp":61,"pain":5,"chiefcomplaint":"weakness and confusion","ground_truth":"HIGH_RISK"}
{"temperature":98.7,"heartrate":82,"resprate":16,"o2sat":98,"sbp":122,"dbp":79,"pain":2,"chiefcomplaint":"minor cut on finger","ground_truth":"LOW_RISK"}
EOF
```

Tool 2: Risk Prediction & Tests Recommendation Module

Context-Aware Outputs:
Diagnostic tests are prioritized based on historical case similarity and relevance.

Improved Interpretability:
Clinicians can see why specific tests were recommended through case-based evidence.


```
[{'test': 'EKG', 'support_from_similar_cases': 2, 'priority': 'Routine'},
 {'test': 'Blood culture',
  'support_from_similar_cases': 2,
  'priority': 'Routine'},
 {'test': 'CBC', 'support_from_similar_cases': 2, 'priority': 'Routine'},
 {'test': 'Serum lactate',
  'support_from_similar_cases': 2,
  'priority': 'Routine'},
 {'test': 'BMP', 'support_from_similar_cases': 1, 'priority': 'Routine'},
 {'test': 'Type & screen',
  'support_from_similar_cases': 1,
  'priority': 'Routine'}]
```


	subject_id	stay_id	chiefcomplaint	temperature	heartrate	resprate	o2sat	sbp	dbp	pain	acuity	risk_prob	risk_label	tests_recommended
0	11482036	36735480	ALTERED MENTAL STATUS	97.8	40.0	18.0	97.0	148.0	20.0	0.0	3.0	0.962981	1	EKG|Routine; Serum lactate|Routine; Blood cult...
1	18001424	32228682	COUGH/ILI	97.0	76.0	22.0	93.0	162.0	20.0	0.0	3.0	0.427643	1	Chest X-ray|Routine
2	18784631	37865787	ABD PAIN, ABDOMINAL DISTENTION	97.0	77.0	18.0	98.0	115.0	20.0	8.0	3.0	0.450036	1	CBC|Routine; CMP|Routine
3	14841168	32335445	ALTERED MENTAL STATUS	98.1	80.0	20.0	100.0	97.0	20.0	10.0	3.0	0.925758	1	EKG|Routine; BMP|Routine
4	11834402	33153219	DIARRHEA, FATIGUE	101.9	84.0	14.0	95.0	97.0	20.0	0.0	3.0	0.529599	1	CBC|Routine; Serum lactate|Urgent

Tool 2: Risk Prediction & Tests Recommendation Module

Model	AUROC	Precision	Recall (Sensitivity)	Specificity	Accuracy	Coverage (Must)	Coverage (Should)	Over-Ordering	Priority Hit	Avg Tests/Case
Model 1 – Random Forest + BioMistral-7B	0.81	0.64	0.69	0.77	0.74	0.55	0.48	0.09	0.92	3.2
Model 2 – XGBoost + Llama-3.1 Instruct	0.84	0.7	0.75	0.74	0.77	0.55	0.51	0.07	0.95	3.6
Model 3 – CatBoost + Qwen 2.5 Guarded	0.85	0.65	0.79	0.73	0.76	0.62	0.53	0.03	1	2.48
Model 4 – CatBoost + Retrieval Memory (RAG)	0.85	0.65	0.79	0.73	0.76	1	0.72	0.45	0.86	4.78

Metric	Description / Purpose
Accuracy	Proportion of correct predictions overall.
Precision	Fraction of predicted high-risk cases that were true high-risk.
Recall (Sensitivity)	Fraction of true high-risk cases correctly identified.
F1-Score	Harmonic mean of precision and recall for balanced assessment.
AUROC / PR-AUC	Measures of discriminative ability under class imbalance.
Confusion Matrix	Visualization of correct vs. misclassified outcomes for interpretability.

Metric	Purpose
Coverage (Must / Should)	Percentage of essential and contextually relevant tests returned.
Over-Ordering Ratio	Rate of unnecessary or redundant test suggestions.
Priority Hit Rate	Percentage of high-risk cases receiving at least one critical test.
Schema Compliance	Validation of JSON output structure for safety and consistency.

Tool 3: Symptom-Diagnosis Module

The dataset used in this project comes from a medical symptoms dataset from Hugging Face.
It includes 132 symptoms as features and a target column called “prognosis”, which represents the disease.
Each symptom is binary — ‘1’ indicates presence, and ‘0’ indicates absence.
The dataset covers 40 different diseases, making it well-balanced for multi-class classification.

Dataset attributes:

- Fatigue (binary) – tiredness or low energy
- Loss of Appetite (binary) – reduced desire to eat
- Fever (binary) – elevated body temperature
- Cough (binary) – presence of dry or productive cough
- Prognosis (target label) – represents the predicted disease (e.g., Fungal infection, Allergy, Malaria, etc.)

palpitations	painful_wal	pus_filled_p	blackheads	scurring	skin_peelin	silver_like_	small_dent	inflammatc	blister	red_sore_a	yellow_crus	prognosis
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	0	0	0	0	0	0	0	Urinary tract infection
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	0	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	0	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	0	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	0	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	1	1	1	1	0	0	0	Psoriasis
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	0	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	0	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	0	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	1	1	1	Impetigo
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection
0	0	0	0	0	0	0	0	0	0	0	0	Fungal infection

Tool 3: Symptom-Diagnosis Module

Model Architecture and Algorithms

Support Vector Classifier (SVC) – achieved 87.7% accuracy, performed best overall by finding an optimal hyperplane to separate disease classes.
Gaussian Naïve Bayes (NB) – a probabilistic model, fast but less accurate here with ~44.9% accuracy.
Random Forest (RF) – an ensemble of decision trees achieving ~74% accuracy.

- After evaluating individual models, a Voting Classifier was created to combine predictions from all three.

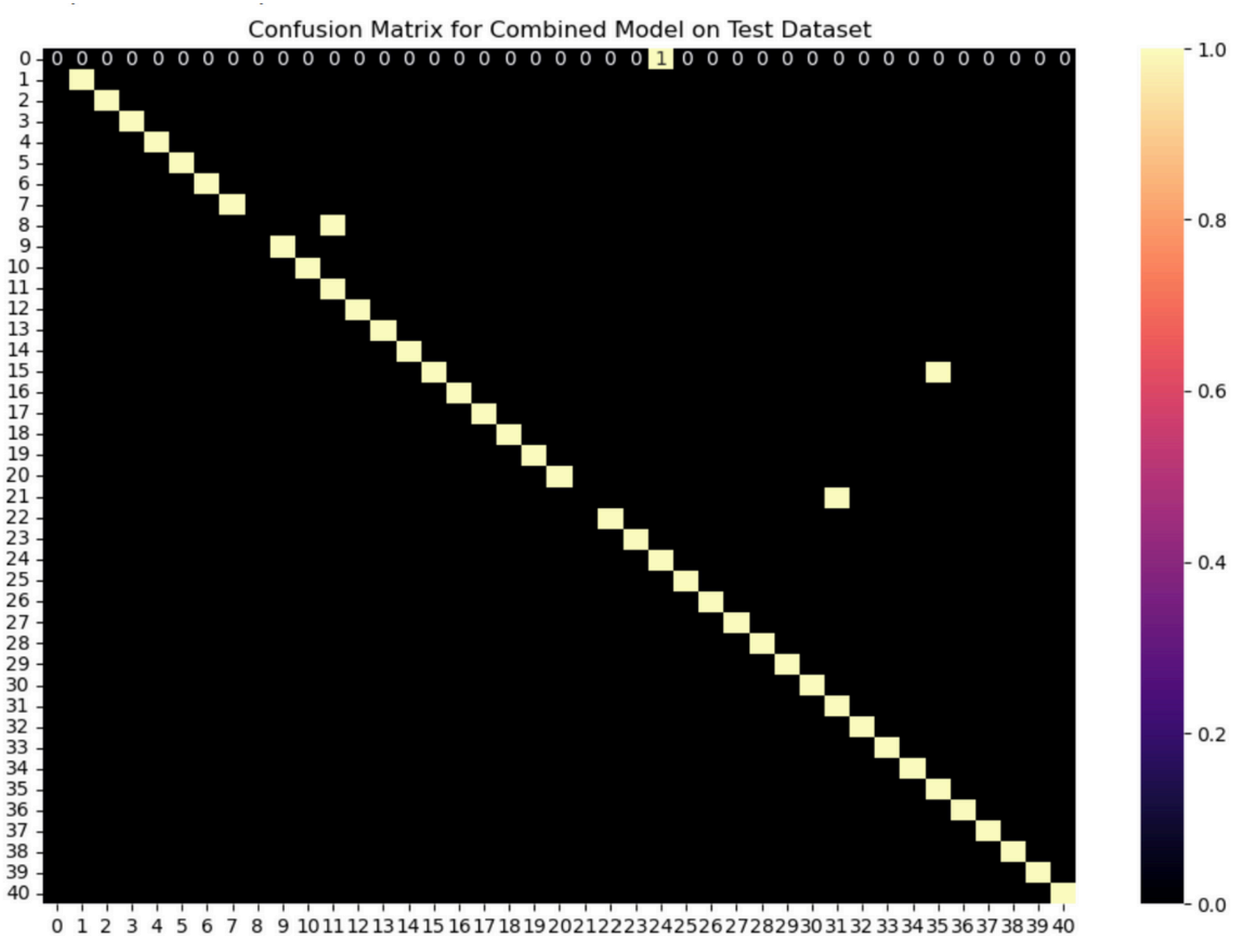
```
SVC test accuracy: 0.8770325203252033
NB test accuracy: 0.4491869918699187
RF test accuracy: 0.7408536585365854
```

Fitting the model on whole data and validating on the Test dataset

Tool 3: Symptom-Diagnosis Module

Model Evaluation and Confusion Matrix Analysis

- The final combined model was trained on the full dataset and validated on the test set.
- The Confusion Matrix shown here illustrates near-perfect alignment between predicted and actual disease classes — most values lie on the diagonal, showing highly accurate predictions.
- This visualization confirms that the model has strong generalization and minimal misclassification across all 40 disease categories.



Tool 4: MedQA Module

Datasets

MedQA (USMLE-style): Multiple-choice, exam-like medical knowledge questions spanning core disciplines

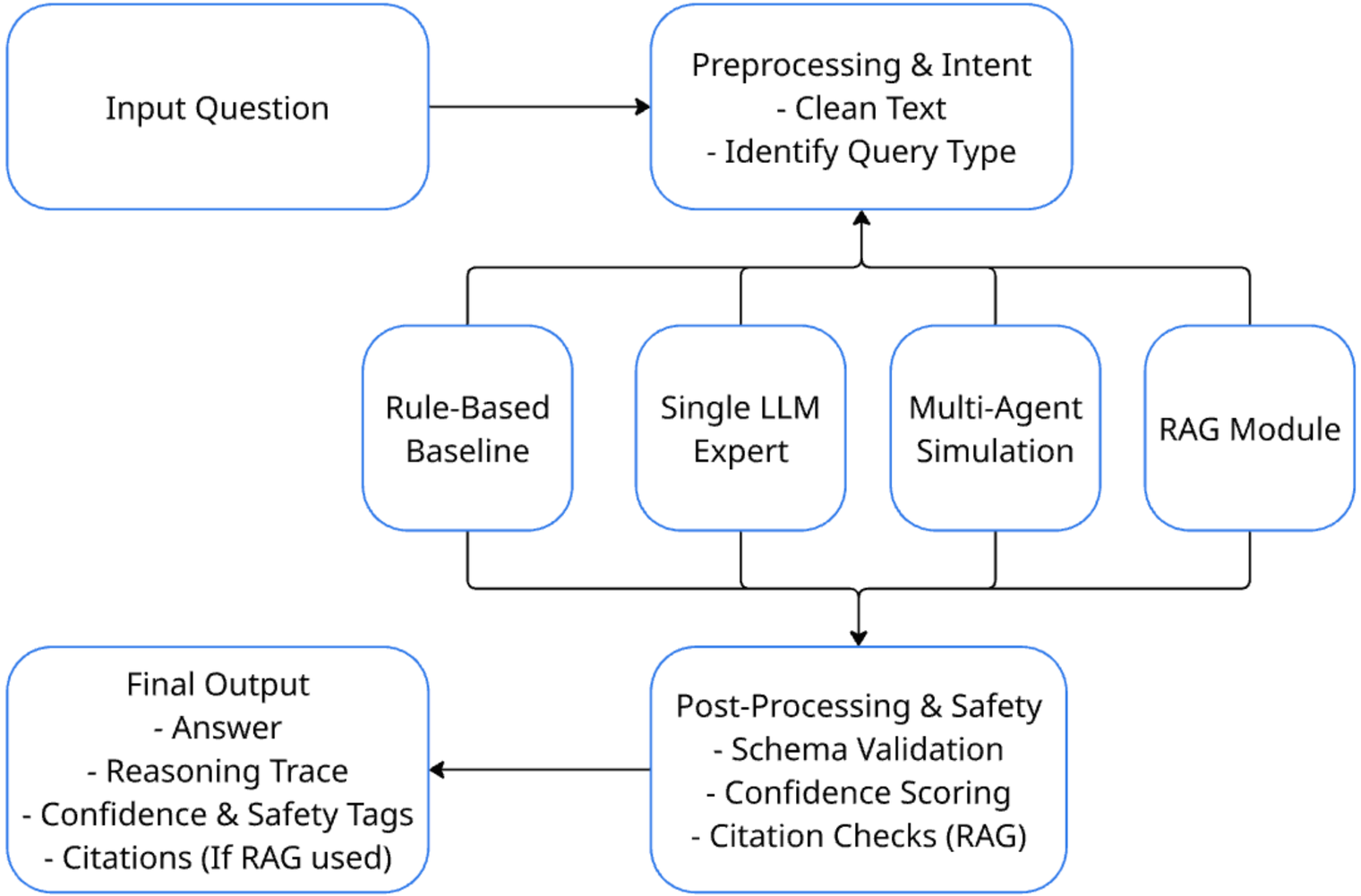
PubMedQA: Biomedical Q&A pairs derived from PubMed abstracts with short, evidence-oriented answers.

Models

Method	Strength	Best use	Limits	Typical latency*
Rule-Based	Deterministic, instant	Definitions/patterns	Narrow coverage	~instant
Single-LLM	Fluent, adaptable	General QA	Calibration, hallucination risk	low–moderate
Multi-Agent	Conservative consensus	Ambiguous/high-stakes	Higher cost/latency	higher

Tool 4: MedQA Module

Workflow:



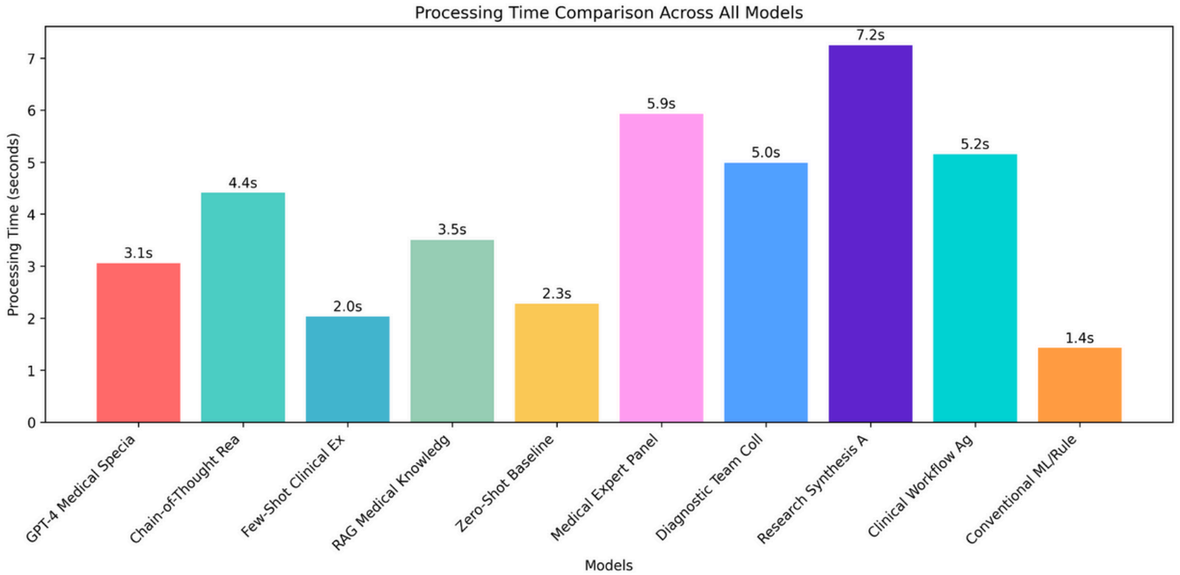
Tool 4: MedQA Module

Example:


```
{
  "question": "what are symptoms of anxiety disorder?",
  "qid": "user_demo_20251103_192302",
  "source": "user_input",
  "timestamp": "2025-11-03T19:23:16.479487",
  "results": {
    "gpt4_medical": {
      "method": "GPT-4 Medical Specialist",
      "answer": "Medical Specialist Analysis: Anxiety disorders are complex mental health conditions characterized by persistent, excess
      "confidence": "88.8%",
      "processing_time": 3.0567172580914055,
      "reasoning": "Specialized medical prompting with comprehensive clinical knowledge",
      "safety_score": 0.917790784365585,
      "timestamp": "2025-11-03T19:23:16.479455",
      "prompting_details": {
        "medical_specialization": {
          "processing_time": 0.904071479316139
        },
        "clinical_knowledge": {
          "processing_time": 0.7410514349245163
        },
        "comprehensive_analysis": {
          "processing_time": 0.761691789185107
        }
      }
    },
```


📊 MODEL COMPARISON TABLE

Model	Type	Time(s)	Confidence	Safety
Zero-Shot Baseline	Single LLM	2.4	76.3%	85.0%
Few-Shot Clinical Exampl	Single LLM	2.9	87.9%	91.8%
GPT-4 Medical Specialist	Single LLM	2.6	84.4%	84.2%
Chain-of-Thought Reasoni	Single LLM	4.6	86.0%	91.1%
RAG Medical Knowledge	Single LLM	4.2	93.5%	89.9%
Medical Expert Panel	Multi-Agent	5.4	94.3%	89.1%
Diagnostic Team Collabor	Multi-Agent	5.5	94.2%	82.1%
Research Synthesis Agent	Multi-Agent	7.7	94.4%	96.9%
Clinical Workflow Agents	Multi-Agent	5.7	88.4%	86.6%
Conventional ML/Rules	Conventional	1.9	72.8%	86.8%



FUTURE WORK

1. Model Deployment

- Deploy all fine-tuned models on Vertex AI.
- Expose secure model endpoints for downstream use.

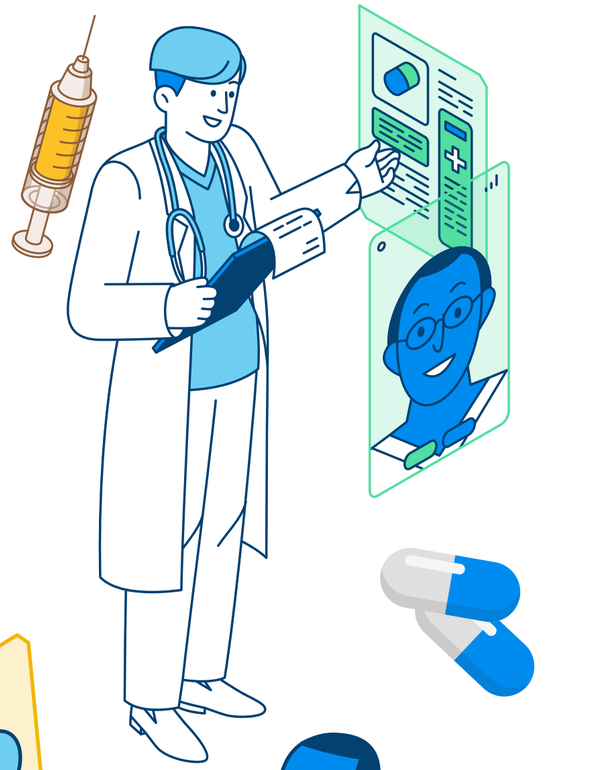
2. Agentic Workflow Development

- Connect summarizer, diagnosis, Q&A, and risk modules.
- Orchestrate through the LangGraph framework.

3. Heterogeneous RAG System

- Integrate multiple datasets into a single retrieval pipeline.
- BigQuery for structured patient data.
- Vertex AI Vector Index for unstructured notes and documents.

4. Observability & Monitoring

- Implement LangSmith for evaluations, traces, and error analysis.
- Use Cloud Logging & Monitoring for system health

THANK YOU!